

TASK - 8

Objective: Understand the role of VPNs in protecting privacy and secure communication.

Step 1: Choose a Reputable Free VPN and Sign Up

VPN Chosen: **Proton VPN**

Reason for Choice: Proton VPN, developed by the team behind Proton Mail, has a strong reputation for privacy.

Its free version has:

No data logging. (Crucial for privacy)

No data caps. (Unlimited bandwidth)

No ads.

And other vpn services like OpenVPN , Windscribe..etc .

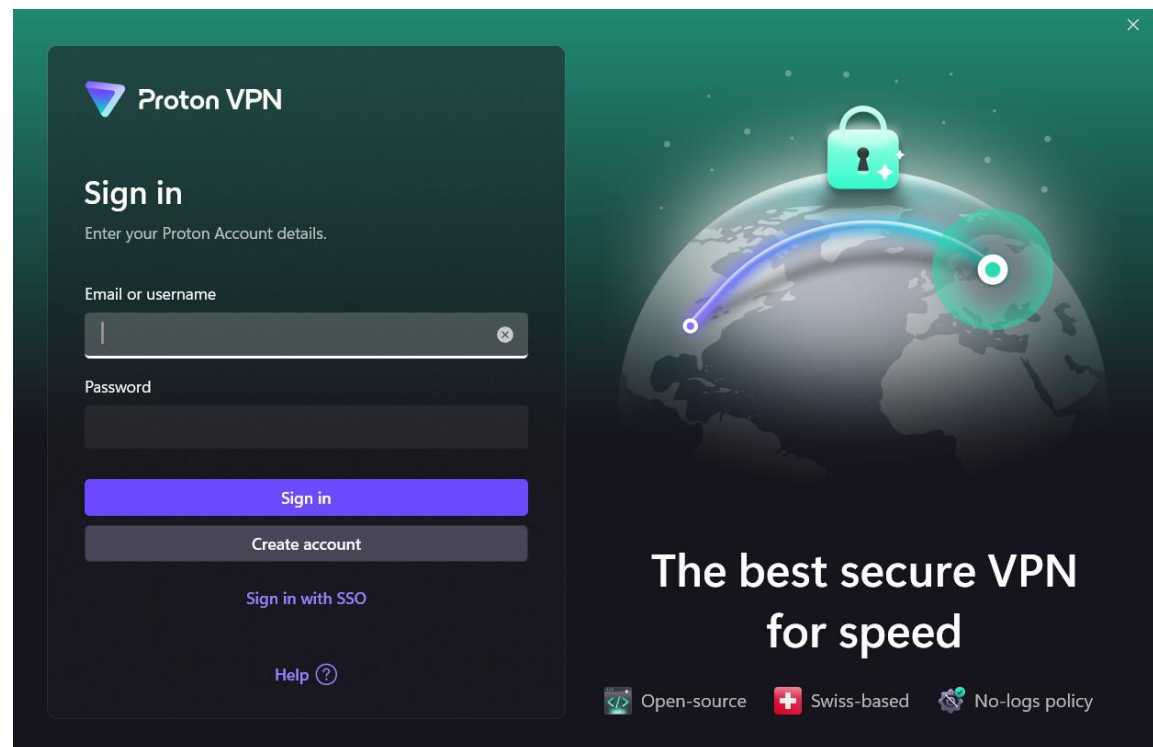
Installation guide :

Go to the official Proton VPN website: <https://protonvpn.com>

Click on "Get Proton VPN" or "Free Download."

Select the Free plan.

Sign up with an email address and a secure password.



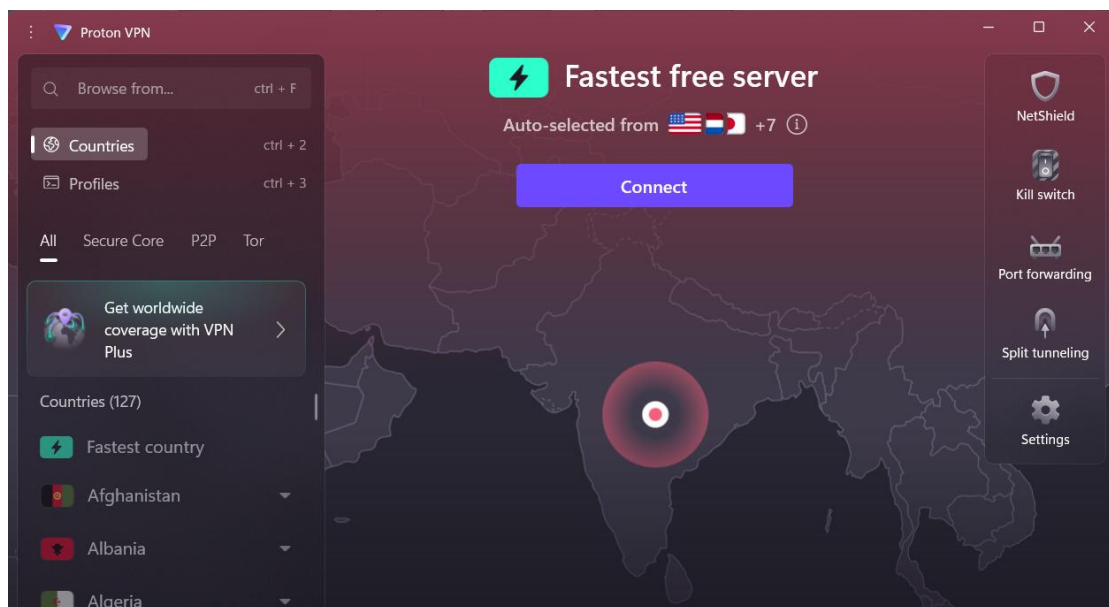
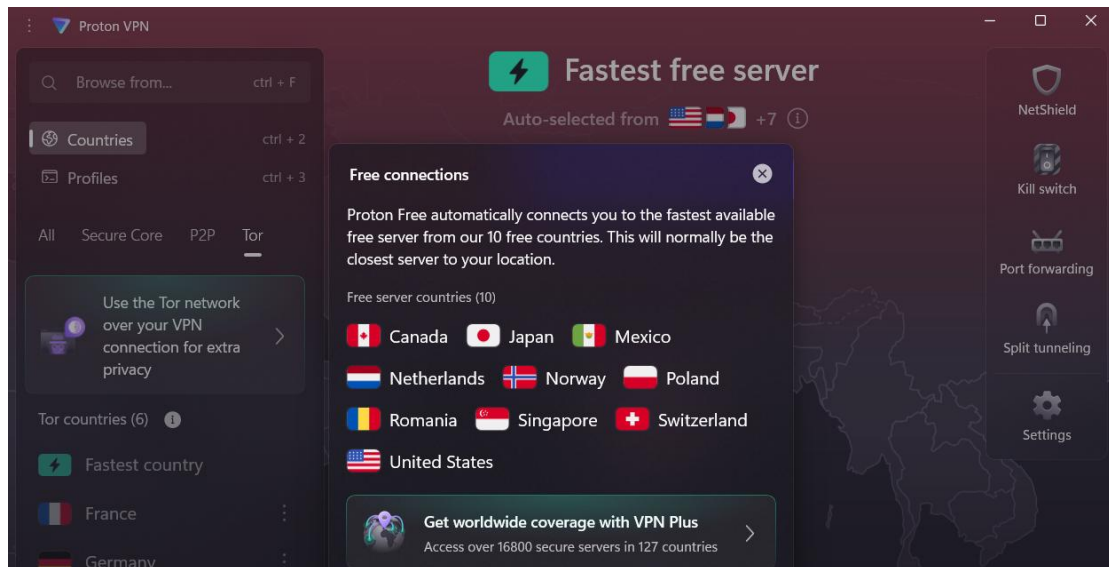
Step 2 : Connect to a VPN Server

Log in with the Proton Account you just created.

You will see a list of available servers. The free servers are marked with a “Free” .

As recommended, choose the server closest to your physical location for the best possible speed.

Click on the server name and then click "Connect."



Before connecting to VPN first check you IP Address for this follow the steps:

- Open your web browser.
- Go to <https://whatismyipaddress.com>
- You will find your public IP Address in it.

What Is My IP?

My IPv4: [27.116.16.154](#)

My IPv6: Not Detected

MY IP LOCATION INFORMATION

City: Hyderabad

State/Region: Telangana

Postal Code: 500018

Country: India

ISP: Vainavi Industries Ltd.

ASN: [45582](#)

Time Zone: UTC +05:30

View larger map

India

MADHYA PRADESH

CHHATTISGARH

Nagpur

Amravati

Bhilai

Raipur

Google

Keyboard shortcuts Map data ©2025 Google Terms

Now the connect to the VPN

Proton VPN

Browse from... ctrl + F

Countries ctrl + 2

Profiles ctrl + 3

All Secure Core P2P Tor

Get worldwide coverage with VPN Plus

Countries (127)

Fastest country

Afghanistan

Albania

Algeria

Angola

Argentina

Fastest free server

Romania - RO-FREE#26

Change server Disconnect

Protected • 18 sec

VPN IP: 185.252.220.147

Server load: 85%

Current traffic (KB/s): 35 KB/s (down), 14 KB/s (up)

Protocol: WireGuard (UDP)

Total traffic: 342 KB

NetShield

Kill switch

Port forwarding

Split tunneling

Settings

Step 3: Verify Your IP Address Has Changed :

Again open browser go to <https://whatismyipaddress.com>

Check the IP Address in the system.

Before connecting the VPN, note down your original IP address and general location.

After connecting the VPN, refresh the whatismyipaddress.com page.

You will see a completely different IP address and location, corresponding to the VPN server you connected.



Step 4: Browse a Website to Confirm Traffic is Encrypted

While still connected to the VPN, visit any website, for example, <https://www.wikipedia.org>.

While you can't see the encryption directly in your browser, but the VPN client uses an encrypted "tunnel" between your device and the VPN server.

Step 5: Disconnect VPN and Compare Browsing Speed and IP

For this go to a speed test website like <https://www.speedtest.net>.

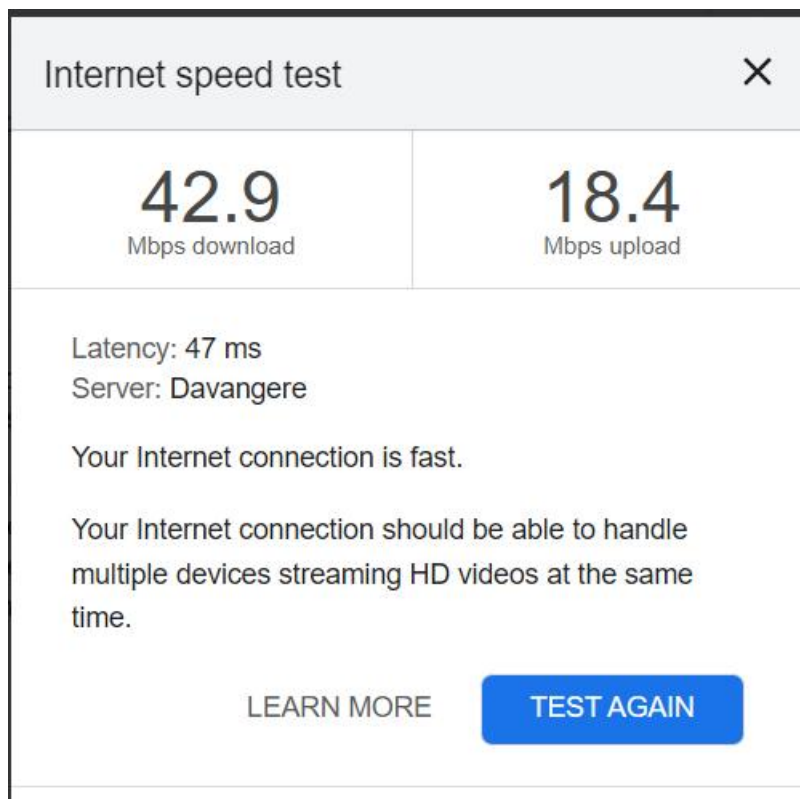
Run the speed test when using vpn and note the download/upload speeds.

Go back to the Proton VPN client and click "Disconnect."

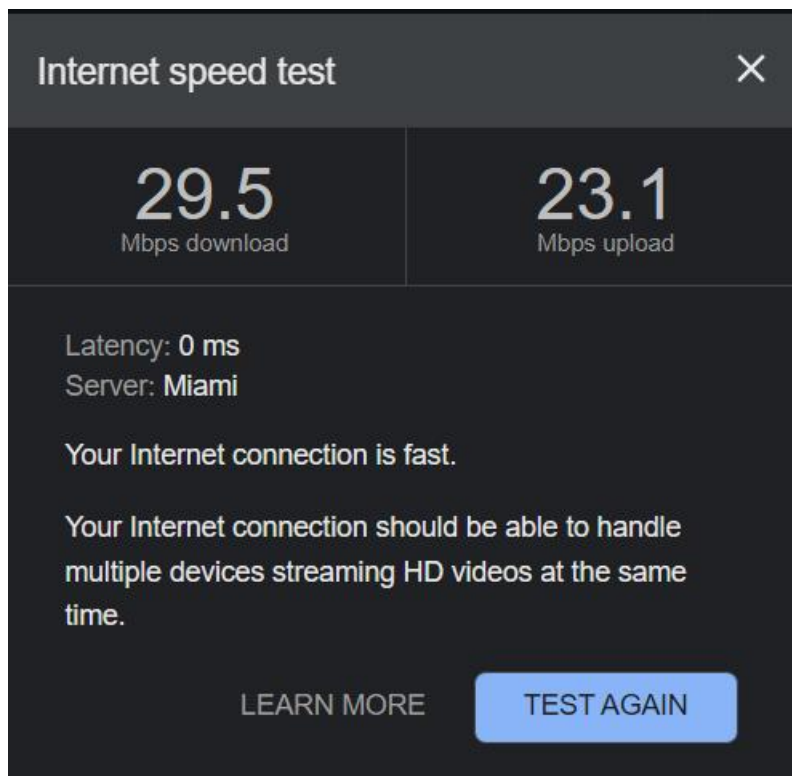
Refresh the whatismyipaddress.com page again. You will see that your IP address has reverted to your original, real IP address.

Run the speed test without the VPN and note the download/upload speeds.

Speed test before connecting VPN :



Speed test after connecting VPN :



Step 6: Research VPN Encryption and Privacy Features

For Proton VPN, it use AES-256 encryption. This is the same military-grade encryption standard used by governments and security experts worldwide. It is currently considered unbreakable by brute force.

Protocols: They use secure protocols like OpenVPN and WireGuard to create the encrypted tunnel.

Privacy Features:

No-Logs Policy: A true no-logs policy means the VPN provider does not record your online activity, connection timestamps, or original IP address.

Kill Switch: This feature automatically blocks all internet traffic if the VPN connection drops unexpectedly, preventing your real IP from being exposed.

DNS Leak Protection: Ensures that your DNS queries (which translate website names to IP addresses) are also routed through the VPN tunnel and not your ISP's servers.

Step 8: Summary on VPN Benefits and Limitations

Benefits of Using a VPN:

Enhanced Privacy: Hides your IP address and browsing activity from your Internet Service Provider (ISP) and other local network eavesdroppers, especially on public Wi-Fi.

Improved Security on Public Wi-Fi: Encrypts your data, protecting you from hackers on unsecured networks like those in cafes, airports, and hotels.

Bypass Geographic Restrictions: Allows you to access content and services that may be restricted in your area.